H ATTENTION PATTERNS AND PREFIX-MATCHING SCORE

Across a variety of tasks, the heads with highest causal effect have a consistent attention pattern where the attention weights on few-shot ICL prompts are the strongest on the output tokens of each in-context example. Here we show this pattern for GPT-J on 4 additional tasks (Figure 23, Figure 24), which match the patterns shown in the main paper (Figure 3b). This is similar to the attention pattern that might be expected of "induction heads", which has previously been shown to arise when a prompt contains some repeated structure (Elhage et al., 2021; Olsson et al., 2022).

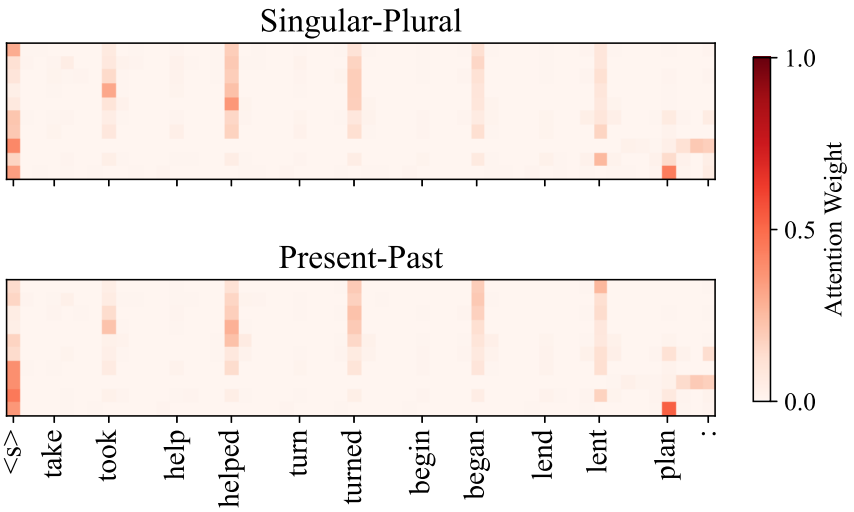

Figure 23: Attention weight visualizations for the singular-plural and present-past tasks for the attention heads with the top 10 average indirect effects in GPT-J. Across tasks, the attention weights are consistently the strongest on the output tokens of each exemplar.

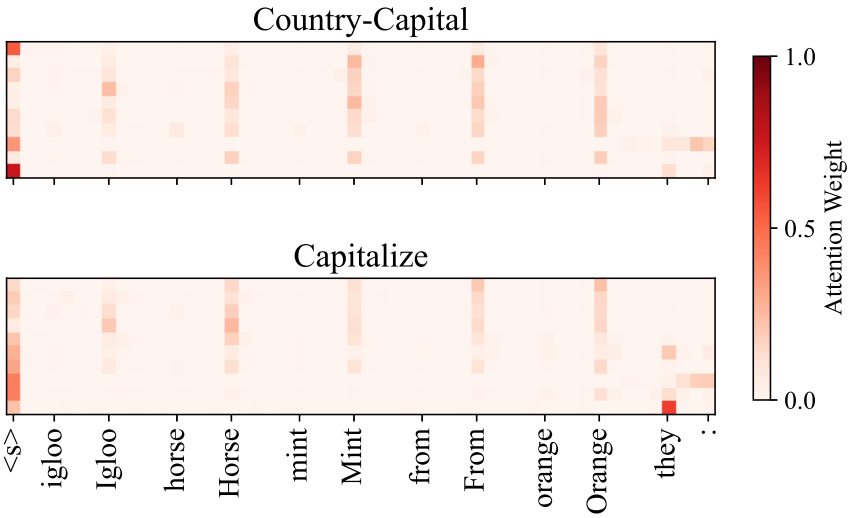

Figure 24: Attention weight visualizations for the country-capital and capitalize tasks for the attention heads with the top 10 average indirect effects in GPT-J. Across tasks, the attention weights are consistently the strongest on the output tokens of each exemplar.

To further investigate whether the heads identified via causal mediation analysis are "induction heads", we compute the prefix-matching score for each head in GPT-J. We follow the same procedure as described in (Olsson et al., 2022; Wang et al., 2022a), which computes the prefix-matching score as the average attention weight on a token B when given a sequence of the form $[A, B, \ldots, A]$. This is

measured on sequences of repeated random tokens. We do this for each head in GPT-J with results shown in Figure 25b.

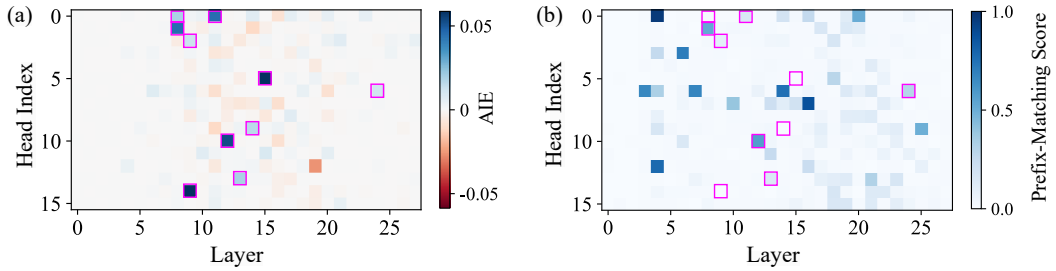

Figure 25: (a) Average Indirect Effect (AIE) per attention head for GPT-J. (b) Prefix-matching score per attention head for GPT-J. For both (a) and (b), we highlight in pink the top 10 heads by AIE. There are three heads that have both a relatively high AIE and prefix-matching score (Layer-Head Index = 8-1, 12-10, and 24-6). There are also several heads with high AIE that do not have a high prefix-matching score, and vice-versa.

We find that three of the heads out of those with the top 10 highest AIEs (Figure 25) also have high prefix-matching scores. In terms of "Layer-Head Index", these are heads 8-1, 12-10, and 24-6, with prefix-matching scores of 0.49, 0.56, and 0.31, respectively.

While (Elhage et al., 2021; Olsson et al., 2022) show that induction heads play a critical role in copying forward previously seen tokens, our results show that they are also among the set of heads, $\mathcal{A}$, that have the highest AIE when resolving few-shot ICL prompts.

There are several other heads we identified with relatively high causal effect that have the same attention pattern activation on few-shot ICL prompts, but do not produce the same "induction" attention pattern on sequences of random repeated tokens.

This suggests that while induction heads play a role in the formation of function vectors, there are other heads that also contribute relevant information that may not be induction heads of the type observed by (Elhage et al., 2021; Olsson et al., 2022).

I DECODING VOCABULARY EVALUATION

Table 19: Additional tasks and the top 5 vocabulary tokens of their decoded FV. Across a variety of outputs, most encodings are aligned to the task they were extracted from.

Task t	**Tokens in the distribution** $D(v_t)$ **in order of decreasing probability**
Capitalize First Letter	‘CN’, ‘DR’, ‘RR’, ‘ Ct’, ‘Ct’
Country-Currency	‘ Japanese’, ‘ Chinese’, ‘ Arabic’, ‘ Russian’, ‘ American’
English-German	‘ âĶĬ’, ‘ ËĪ’, ‘ è’, ‘actual’, ‘ ç¥l’
English-Spanish	‘ âĶĬ’, ‘ è’, ‘ ç¥l’, ‘ masc’, ‘operator’
Landmark-Country	‘ Germany’, ‘ Japan’, ‘ Netherlands’, ‘ Italy’, ‘ Spain’
Lowercase First Letter	‘dr’, ‘ nr’, ‘ lc’, ‘ mc’, ‘ mr’
National Parks	‘ Connecticut’, ‘ California’, ‘ Wisconsin’, ‘ Netherlands’, ‘ Pennsylvania’
Park-Country	‘ Netherlands’, ‘ Germany’, ‘ Japan’, ‘ Italy’, ‘ Mexico’
Person-Sport	‘ basketball’, ‘ football’, ‘ soccer’, ‘ baseball’, ‘ tennis’
Product-Company	‘ Microsoft’, ‘ Motorola’, ‘ Samsung’, ‘ Disney’, ‘ IBM’
Sentiment	‘ positive’, ‘ negative’, ‘positive’, ‘negative’, ‘ neutral’
Synonym	‘ edible’, ‘ adjective’, ‘ noun’, ‘ slang’, ‘ caster’

Here, we present more results on the evaluation of decoding vocabularies of FVs, over additional datasets. Across the tasks, we affirm that output spaces seem to be frequently encoded in the FVs, as can be seen in Table 19. In particular, in cases such as sentiment that follow a rigid pattern, the tokens referring to the output distribution for sentiment is well encoded. On the other hand, some tasks like language translation do not have output spaces well-encoded in the FVs.